



| Stator Resistance Measurement |           |          |                  |
|-------------------------------|-----------|----------|------------------|
| S.No.                         | V (volts) | I (amps) | Rdc=V/I $\Omega$ |
|                               |           |          |                  |
| Average                       |           | Rdc      |                  |

**SAMPLE CALCULATION (Set No.    )**

Radius of brake drum R = .....m

Synchronous speed  $N_s$  = .....rpm

Input Power  $W = W_1 + W_2 =$  \_\_\_\_\_ W

Power factor  $\cos\phi = \cos\left[\tan^{-1} \frac{\sqrt{3} \times (W_1 - W_2)}{(W_1 + W_2)}\right] =$

Percentage slip,  $s = \frac{N_s - N}{N_s} \times 100 =$  \_\_\_\_\_ %

Torque  $T = R \times (S_1 - S_2) \times g =$  \_\_\_\_\_ N-m

Output power  $= \frac{2\pi NT}{60} =$  \_\_\_\_\_ W

Efficiency  $= \frac{\text{Output}}{\text{Input}} \times 100 =$  \_\_\_\_\_ %